Homework #7 — Confidence Intervals and Normal Distribution (15 pts)

MATH 240 — Introduction to Statistics, Fall 2026

Due Date: Monday, 11/9 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as one .pdf file.
- Responses can be hand-written or typed for the textbook problems.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- For all questions involving code, please provide your code along with the output.
- Refer to syllabus for late work deductions.

Background

We will now focus on confidence intervals and the normal distribution, which will provide us with more interpretations about sample statistics from an experiment aside from just point estimates.

All problems in this week's homework are from the textbook. For questions that do require R code, please write/type down the R code in your work.

Textbook Problems (1.5-3 pts each)

Complete the following problems:

1. Chapter 12: Q1, Q2, Q6, Q8
2. Chapter 13: Q2, Q4, Q6